



Sex bias in pain management decisions

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In the pursuit of mental and physical health, effective pain management stands as a cornerstone. Here, we examine a potential sex bias in pain management. Leveraging insights from psychological research showing that females' pain is stereotypically judged as less intense than males' pain, we hypothesize that there may be tangible differences in pain management decisions based on patients' sex. Our investigation spans emergency department (ED) datasets from two countries, including discharge notes of patients arriving with pain complaints (N = 21,851). Across these datasets, a consistent sex disparity emerges. Female patients are less likely to be prescribed pain-relief medications compared to males, and this disparity persists even after adjusting for patients' reported pain scores and numerous patient, physician, and ED variables. This disparity extends across medical practitioners, with both male and female physicians prescribing less pain-relief medications to females than to males. Additional analyses reveal that female patients' pain scores are 10% less likely to be recorded by nurses, and female patients spend an additional 30 min in the ED compared to male patients. A controlled experiment employing clinical vignettes reinforces our hypothesis, showing that nurses (N = 109) judge pain of female patients to be less intense than that of males. We argue that the findings reflect an undertreatment of female patients' pain. We discuss the troubling societal and medical implications of females' pain being overlooked and call for policy interventions to ensure equal pain treatment.

decision-making | pain management | sex bias | healthcare disparities

Pain is ubiquitous. In the United States alone, a staggering 50 million individuals report experiencing pain daily (1). Beyond its direct impact on well-being, pain has critical implications for mental and physical health. If not treated appropriately, pain has the potential to develop into chronic suffering, contribute to medication dependency, and lead to increased healthcare expenditures (2–5). Providing adequate pain management is therefore a central mission of the healthcare system. This task is particularly challenging due to the absence of objective pain measurement tools. Healthcare providers must base their pain management decisions on patients' subjective pain reports and their own clinical judgment and are thus susceptible to different types of biases (6–11).

Here, we investigate whether pain management decisions are biased by patients' sex.* Sex and gender biases have been documented across diverse domains, spanning medical decisions (12), financial investment (13), and scientific and professional communication (14, 15). In the context of pain, laypeople tend to judge females' pain as less intense than that of males (16, 17). For example, when participants watch videos of patients describing their chronic pain, they perceive the pain of female patients as lower compared to that of male patients (17). Recent psychological research suggests that the biased perception of pain stems from a "gender–pain exaggeration bias": Women are believed to report their pain in an exaggerated manner compared to men. If a man and a woman report the same pain score, observers tend to judge the actual pain experienced by the woman as lower than the pain experienced by the man (16). The gender–pain exaggeration bias has been shown to stem from laypeople's perception of women as emotional, dramatic, and even hysterical (16, 18, 19). Women's pain reports face more skepticism than men's and tend to be seen as less authentic (18–20). If healthcare providers hold similar stereotypical views [which is yet unclear (21–24)], then they too may believe that women's pain reports are inflated compared to men's. Such gendered pain perception may affect pain management decisions, resulting in less pain treatment for women.

Do female patients receive less pain treatment than males for similar complaints? Although some have made this claim, a recent review of the literature on sex differences in pain concludes

Significance

Pain treatment must be provided adequately and impartially. Here, we conduct an analysis of archival datasets of pain management decisions as well as a controlled experiment, testing whether pain management differs by patients' sex. We present robust evidence showing that physicians' and nurses' pain management decisions in emergency departments disfavor female patients compared to male patients. Notably, female patients are less likely than males to be prescribed pain-relief medications for the same complaints. We argue that female patients receive less pain treatment than they should, which may adversely impact their health. The findings underscore the critical need to address psychological biases in healthcare settings to ensure fair and efficient treatment for all.

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*We investigate the effects of patients' sex rather than gender because gender identity was not recorded in our databases. Some previous studies have examined the effects of patients' sex, whereas others have looked at gender.

that “anecdotal reports fuel assertions of disparities in pain treatment based on sex” (20). The few studies that tested this question have failed to provide a consistent conclusion. In particular, two studies found that female patients were just as likely as male patients to receive an analgesic prescription but were less likely to receive opioid analgesics (25, 26). Other studies found no disparities in pain management in simulated lab settings (27) or emergency department (ED) and prehospital settings (28–30). One study even reported greater rates of opioid administration to female patients in the ED (29). However, most of the studies investigating sex bias in pain treatment consisted of relatively small cohorts and suffered from methodological limitations, often not controlling for potentially confounding demographic, medical, and healthcare provider variables.

The main goal of the present work is to conduct a rigorous examination of pain management decisions by patients’ sex. We do so by analyzing electronic health records (EHRs) from Israeli and American healthcare systems. Our main analysis tests whether female patients are less likely than males to receive prescriptions of pain relief medication when discharged from the ED, controlling for pain complaints and numerous other variables. Additional analyses examine whether for female patients, the likelihood of nurses recording the pain score is lower, and the length of stay at the ED is longer, both of which have been associated with racial bias (31, 32). We also explore the potential effect of patient–physician sex concordance. Sex concordance has been found to reduce some types of bias in other medical areas (33, 34). Yet, since both male and female physicians may hold the same stereotypes regarding reported pain, they may also exhibit the same-sex bias in treating pain. We complement the archival data analyses with a controlled experiment testing the suggested psychological mechanism of biased stereotypical pain perception among healthcare providers.

Results

Study 1. Study 1 tested our hypothesis that female patients receive less pain treatment than male patients. For this purpose, we collected an EHR dataset of patient discharge notes from EDs in two campuses of the Hadassah-Hebrew University Medical Center in Israel ($N = 17,576$). For each patient, the data included their and the physician’s demographic variables, the medications prescribed upon discharge, the patient’s pain score (reported on a visual analog scale [VAS] from 0 to 10), and variables related to the ED setting (*SI Appendix, Table S1 A and B*).

Our primary outcome of interest, a patient’s likelihood of receiving any analgesic prescription, was lower for female than for male patients [38% vs. 47%, respectively; $\chi^2(1, n = 17,576) = 146.57, P < 0.001, \phi = 0.09$]. This sex disparity was observed in all age groups (Fig. 1*A*), for each pain score (from 0 to 10; Fig. 1*B*) and among male and female physicians (Fig. 1*C*). We also examined the likelihood of receiving an analgesic prescription by sex and by pain level, which were defined according to the WHO pain ladder [i.e., mild, moderate, and severe (35, 36)]. Female patients were less likely than male patients to receive any analgesic prescription for mild pain [VAS 0 to 3; 20% vs. 27%, respectively; $\chi^2(1, n = 817) = 4.30, P = 0.038, \phi = 0.07$], moderate pain [VAS 4 to 6; 33% vs. 41%, respectively; $\chi^2(1, n = 1,800) = 12.29, P < 0.001, \phi = 0.08$], and severe pain [VAS 7 to 10; 50% vs. 59%, respectively; $\chi^2(1, n = 4,208) = 33.28, P < 0.001, \phi = 0.09$]. This sex disparity was also observed for patients with no recorded pain scores [35% vs. 46% for female and male patients, respectively; $\chi^2(1, n = 10,751) = 79.74, P < 0.001, \phi = 0.09$] (Fig. 2*A*).

A similar pattern was observed when examining opioid and non-opioid analgesic prescriptions separately. Compared to male

patients, female patients were less likely to receive an opioid analgesic prescription [19% vs. 25% for female and male patients, respectively; $\chi^2(1, n = 17,576) = 98.54, P < 0.001, \phi = 0.07$]. Specifically, female patients were less likely than male patients to receive an opioid analgesic prescription for moderate pain [14% vs. 19%, respectively; $\chi^2(1, n = 1,800) = 9.11, P = 0.002, \phi = 0.07$] and severe pain [29% vs. 34%, respectively; $\chi^2(1, n = 4,208) = 13.40, P < 0.001, \phi = 0.06$], but not for mild pain [9% vs. 12%, respectively; $\chi^2(1, n = 817) = 0.94, P = 0.331$]. Importantly, female patients were also less likely than male patients to receive a non-opioid prescription [18% vs. 21%, respectively; $\chi^2(1, n = 17,576) = 29.47, P < 0.001, \phi = 0.04$]. Specifically, female patients were less likely to receive a nonopioid analgesic prescription for severe pain [19% vs. 24%, respectively; $\chi^2(1, n = 4,208) = 12.54, P < 0.001, \phi = 0.05$], but not for mild pain [10% vs. 14%, respectively; $\chi^2(1, n = 817) = 3.07, P = 0.080$] and moderate pain [18% vs. 21%, respectively; $\chi^2(1, n = 1,800) = 1.17, P = 0.278$] (Fig. 2*B and C*).

Average pain scores were slightly lower for female patients ($M = 6.64, SD = 2.63$) than for male patients ($M = 6.81, SD = 2.54$), $t(6,824) = -2.75, P = 0.006, d = 0.07$. To control for this and other potential differences in prescription of any type of analgesic between female and male patients, we performed a multivariate linear probability model analysis for those patients with recorded pain scores (Table 1). As predicted, female patients were on average less likely to receive an analgesic prescription than males ($b = -0.09, SE = 0.01, P < 0.001$) (Table 1, Model 1). The sex disparity persisted also after adding controls for patient and ED characteristics ($b = -0.04, SE = 0.01, P = 0.003$) (Table 1, Model 2) and for physician characteristics ($b = -0.03, SE = 0.01, P = 0.004$) (Table 1, Model 3). Finally, we examined the potential effect of patient–physician sex concordance. The interaction between patient sex and physician sex was not a significant predictor of the likelihood of receiving an analgesic prescription ($b = 0.01, SE = 0.02, P = 0.737$) (Table 1, Model 4). The patient’s sex also did not significantly interact with other patient, physician, and ED characteristics in predicting the likelihood of receiving an analgesic prescription (*SI Appendix*).

Of the patients in this study, 1,831 (55% female) attended the ED more than once during the 6 y covered by the data ($N = 4,507$ discharge notes, which are 26% of the discharge notes in our data). To address the potential effect of repeated visits on sex disparity, we performed an additional multivariate regression analysis including only patients with one visit to the ED in our dataset. A similar pattern of sex disparity was observed (*SI Appendix, Table S2*).

The multivariate regression analyses presented thus far included only discharge notes with recorded pain scores (~40% of the total discharge notes). We examined the effect of patient’s sex on analgesic prescription in two additional regression analyses: one of the whole sample (without controlling for pain scores) (*SI Appendix, Table S3A*) and one of patients with no recorded pain score (*SI Appendix, Table S3B*). A similar pattern of sex disparity was observed in these analyses as well.

Finally, we examined two additional measures of pain management in the ED by patients’ sex: triage nurses’ rate of recording patients’ pain scores and patients’ length of stay at the ED. In line with previous findings, both measures were associated with the likelihood of receiving an analgesic prescription, even after controlling for the patients’ sex (see full analyses in *SI Appendix*). Sex disparities were apparent in both. Female patients’ pain scores were less likely to be recorded by triage nurses than males’ pain scores [37% vs. 41%, respectively; $\chi^2(1, n = 17,576) = 41.74, P < 0.001, \phi = 0.05$], even after controlling for patient and ED characteristics ($b = -0.02, SE = 0.01, P = 0.004$) (*SI Appendix, Table S4*). Length of stay at the ED was longer for

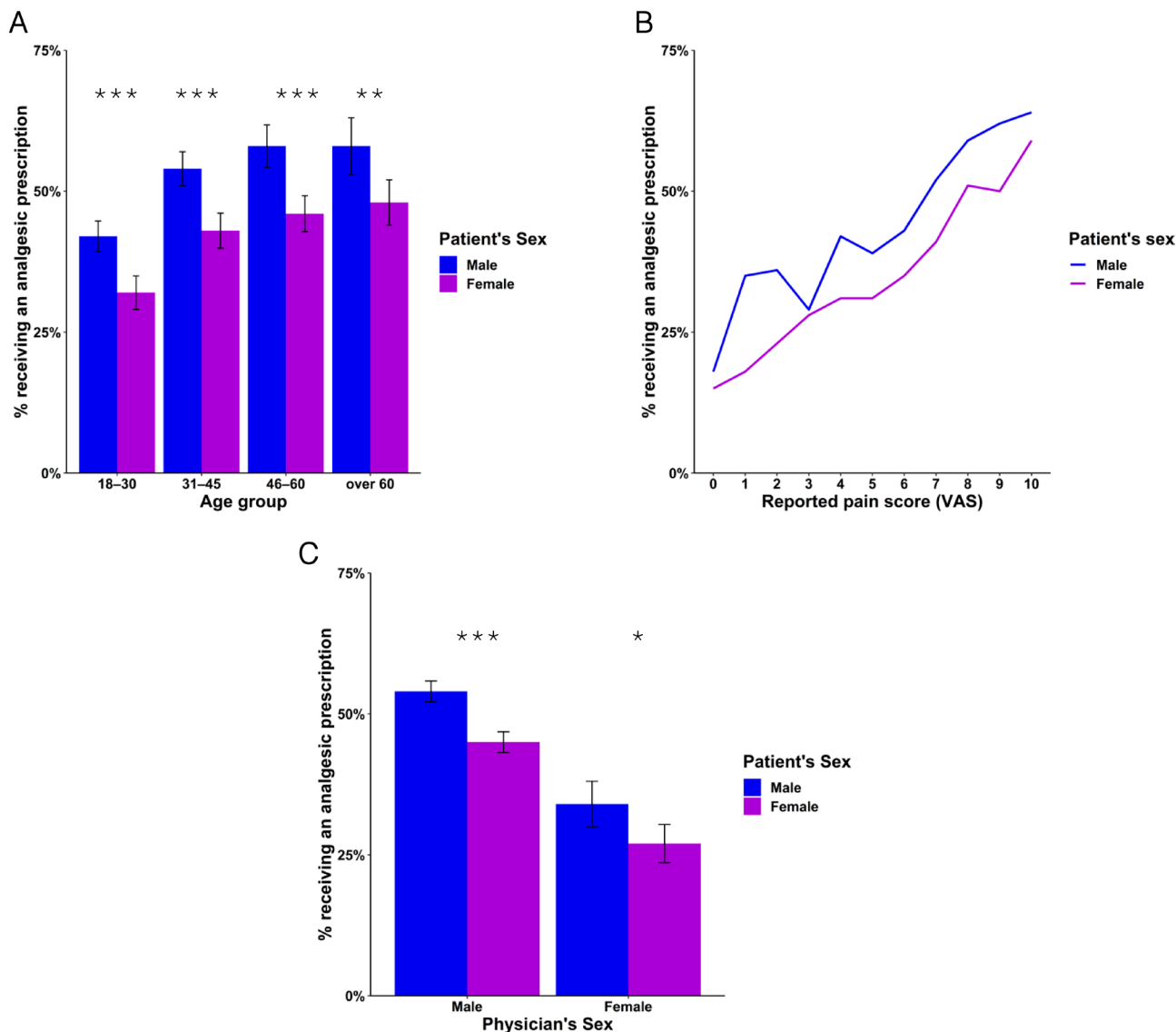


Fig. 1. Likelihood of receiving an analgesic prescription by patient's sex and by (A) age group ($N = 17,576$), (B) reported pain score (VAS) ($N = 6,825$), and (C) physician's sex ($N = 17,576$) (Study 1). Error bars denote 95% CI. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$

female patients' ($M = 4.87$ h, $SD = 2.60$) than for male patients ($M = 4.42$ h, $SD = 2.55$), $t(17,271) = 11.49$, $P < 0.001$, $d = 0.17$, even after controlling for patient, physician, and ED characteristics ($b = 0.16$, $SE = 0.05$, $P = 0.002$) (SI Appendix, Table S5).

Study 2. To examine the generalizability of the findings from Study 1, in Study 2, we obtained a deidentified dataset of discharge notes from the University of Missouri Health Center in Columbia, MO, USA. This dataset was obtained on an aggregate level, precluding us from conducting individual-level data analyses as we did in Study 1. Nevertheless, it allowed us to test whether the findings of Study 1 replicated in a medical system in a different country. As in Study 1, female patients were less likely to receive any type of analgesic prescription than male patients [26% vs. 31%, respectively; $\chi^2(1, n = 4,275) = 12.92$, $P < 0.001$, $\phi = 0.05$]. Female patients were less likely to receive an analgesic prescription for moderate pain levels [20% vs. 28%, respectively; $\chi^2(1, n = 1,006) = 9.28$, $P = 0.002$, $\phi = 0.10$] and for severe pain levels [30% vs. 35%, respectively; $\chi^2(1, n = 2,805) = 7.03$, $P = 0.008$, $\phi = 0.05$] but not for mild pain levels [16% vs. 19%, respectively;

$\chi^2(1, n = 464) = 0.59$, $P = 0.441$] (Fig. 3). Of note, female patients were less likely to receive analgesics even though their average reported pain score (VAS) ($M = 6.91$, $SD = 2.50$) did not differ significantly from male patients ($M = 6.81$, $SD = 2.61$), $t(4,273) = 1.27$, $P = 0.206$, $d = 0.04$.

Study 3. Studies 1 and 2 documented the presence of a sex disparity in healthcare providers' pain management decisions in the field. Study 3 used a controlled experiment to test a potential underlying mechanism, whereby healthcare providers judge female patients' pain as less intense than males' pain. Participants were presented with a brief clinical scenario describing a patient arriving at the ED with back pain, which the patient rated as 9 out of 10. The scenario described the patient as either female or male but was otherwise identical. Participants were asked to judge the intensity of the patient's pain (on a 0 to 100 scale; see SI Appendix for the full scenario). As predicted in a preregistered report, participants rated the patient's pain as less intense when the patient was described as a female ($M = 72.00$, $SD = 26.13$) compared to a male ($M = 80.37$, $SD = 17.15$), $t(107) = 1.98$, $P = 0.051$, $d = 0.38$.

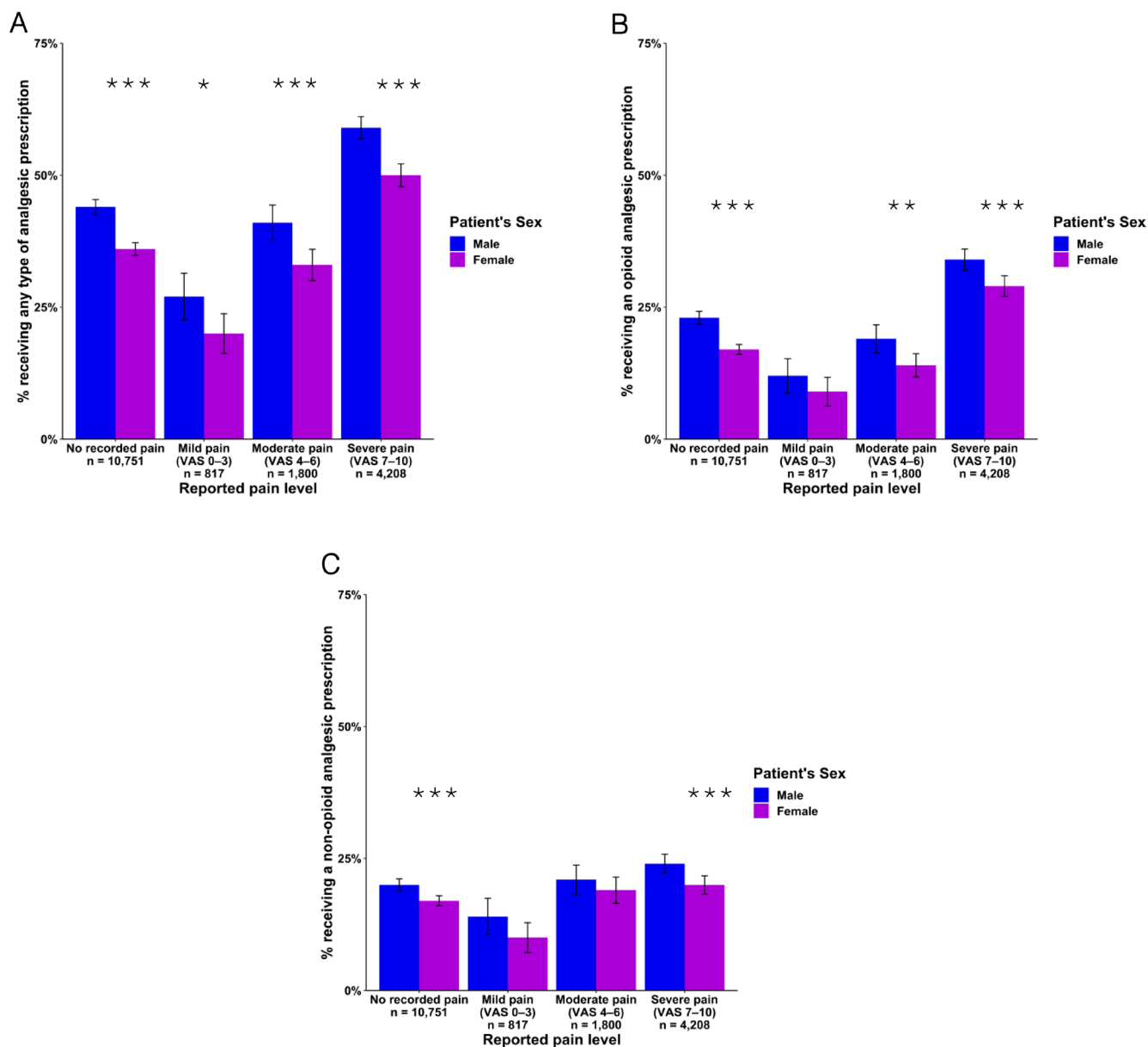


Fig. 2. Likelihood of receiving (A) any type of analgesic prescription, (B) an opioid analgesic prescription, and (C) a nonopioid analgesic prescription by patient's sex and reported pain level (Study 1). Error bars denote 95% CI. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$

Discussion

In 2010, delegates to the International Pain Summit from 130 countries signed the “Declaration of Montreal” calling for recognition of pain management as a basic human right. The declaration’s first article put forward “[t]he right of all people to have access to pain management without discrimination (...). This includes, but is not limited to, discrimination on the basis of age, sex, gender...” (37). The present work reveals a systematic sex-related disparity in pain management: a female patient discharged from the ED is significantly less likely to receive treatment for a pain complaint, compared to a male patient. Datasets from EDs in Israel and the United States, with a total sample size of 21,851 discharge notes, reveal that female patients are less likely to receive a prescription for any type of analgesic medication, both opioids and nonopioids, compared to male patients. Female patients are less likely to receive analgesics for every pain score and at every age group, even when controlling for multiple patient and ED characteristics. Females receive less analgesics from both male and female physicians. Their pain score is less likely to be

recorded by triage nurses, and they stay longer at the ED. Finally, healthcare providers presented with a clinical scenario of a patient in pain, rated the pain as less intense if the patient was said to be female rather than male.

Does the fact that female patients receive less analgesics than males constitute a bias? From a clinical perspective, we argue that it does. Pain management guidelines instruct healthcare providers to prescribe analgesics based on the patient’s reported pain score, regardless of the patient’s sex (35, 36, 38). Yet, we find that even after controlling for the pain score and numerous other factors, pain management decisions depend on the patient’s sex. Does this bias favor or disfavor female patients? It is well known that opioid use carries a risk of addiction, and perhaps underprescription of analgesics is favorable to females (39). Several findings lead us to conclude that female patients are being discriminated against and that females’ pain is being overlooked and undertreated. First, we find that pain scores are recorded by nurses less often for females than for males, although it is recommended that a pain score be recorded for every patient with pain (35, 36). Second, females are

Table 1. Linear regression analysis predicting the likelihood of receiving an analgesic prescription in Study 1, for patients with a recorded pain score

	Model 1	Model 2	Model 3	Model 4
Patient's sex (1 = female)	−0.092*** (0.012)	−0.035** (0.012)	−0.034** (0.012)	−0.027 (0.021)
Reported pain score (VAS)		0.026*** (0.002)	0.026*** (0.002)	0.026*** (0.002)
Patient age group 31 to 45		0.056*** (0.015)	0.056*** (0.015)	0.056*** (0.015)
Patient age group 46 to 60		0.053*** (0.016)	0.052*** (0.016)	0.052*** (0.016)
Patient age group over 60		0.070*** (0.020)	0.069*** (0.020)	0.069*** (0.020)
Patient ethnicity (1 = Jewish)		0.011 (0.013)	0.008 (0.012)	0.008 (0.012)
Patient chronic pain condition		−0.026 (0.040)	−0.029 (0.040)	−0.030 (0.041)
Patient pregnancy		−0.168*** (0.031)	−0.160*** (0.031)	−0.161*** (0.031)
Patient self-referral		−0.015 (0.013)	−0.015 (0.013)	−0.015 (0.013)
Headache		0.013 (0.020)	0.014 (0.019)	0.014 (0.019)
Back pain		0.282*** (0.025)	0.284*** (0.024)	0.284*** (0.024)
Time of discharge (1 = night shift)		−0.062*** (0.018)	−0.058*** (0.014)	−0.058*** (0.014)
Number of patients in ED		0.000 (0.000)	−0.001 (0.001)	−0.001 (0.001)
Type of ED (1 = surgical)		0.142*** (0.025)	0.144*** (0.025)	0.144*** (0.025)
Campus (1 = Mt. Scopus)		−0.020 (0.046)	−0.023 (0.046)	−0.022 (0.046)
Year 2015		−0.016 (0.035)	−0.016 (0.035)	−0.016 (0.035)
Year 2016		−0.029 (0.031)	−0.023 (0.031)	−0.023 (0.031)
Year 2017		−0.011 (0.033)	−0.004 (0.033)	−0.004 (0.033)
Year 2018		0.006 (0.031)	0.015 (0.031)	0.015 (0.031)
Year 2019		0.047 (0.029)	0.054 (0.030)	0.054 (0.030)
Length of stay at ED		−0.008** (0.003)	−0.008** (0.003)	−0.008** (0.003)
Physician's sex (1 = female)			−0.062* (0.030)	−0.058* (0.029)
Physician age			0.001 (0.002)	0.001 (0.002)
Physician ethnicity (1 = Jewish)			0.032 (0.033)	0.032 (0.033)
Resident physician			0.019 (0.030)	0.019 (0.030)
Patient's sex X Physician's sex				0.008 (0.025)
Number of observations	6,825	6,825	6,825	6,825
R-squared	0.008	0.243	0.246	0.246

The table presents the estimates and their SE in brackets.

* $P < 0.05$;

** $P < 0.01$;

*** $P < 0.001$.

also less likely to receive nonopioid analgesics (e.g., ibuprofen), which are safe and nonaddictive. Third, for severe pain levels (i.e., VAS 7 to 10), the guidelines recommend that *all* patients receive an analgesic (35, 36). While we find an overall underprescription of analgesics relative to this recommendation [in line with previous findings (40, 41)], the divergence from the guidelines is greater for female patients than for males, implying less adequate pain management for females. Finally, the results of the clinical scenario study suggest that healthcare providers underestimate the pain reports of females compared to those of males. Going beyond medical guidelines, one could also ask whether biological differences in pain processing may warrant the prescription of less analgesics to females. The current literature seems to suggest that the opposite is more likely to be true. Females seem to be physiologically *more* sensitive to pain (42, 43), implying that they might need more—not less—analgesics than males.

Why are healthcare providers less likely to treat the pain of females than that of males? Previous work has shown that individuals tend to perceive females' pain as less intense than that of males. This perception bias has in turn been explained by a gender–pain exaggeration bias: People view females as more emotional and assume that they overreport their experienced pain compared to males (16, 18, 19). The literature has suggested additional stereotypes that may also explain why female patients receive less pain treatment, such as the perception of females as more capable

of physically tolerating pain than males (19). While other stereotypes may be at play, our findings of bias in healthcare providers' pain perception are in line with the exaggeration mechanism. We note that our data did not allow us to tie biased pain management decisions to stereotypical pain judgments within the same study. Previous work has been able to establish such a connection in the context of racial bias, where a stereotypical perception of Black patients as more physically resilient to pain has been associated with inadequate pain treatment (11). Although both Black and female patients end up receiving less pain treatment, the particular stereotypes underlying this bias seem to differ: Black people are perceived as resilient while females are perceived as exaggerating their pain. Additional mechanisms, beyond stereotypes, may be driving the sex bias. Females and males may communicate their pain differently, in a way that makes male pain seem more deserving of treatment (18, 19, 44). Relatedly, females may be less inclined to request pain relief medication, in line with the general finding that “women don't ask” (45). Future research should better differentiate the mechanisms at play, as understanding them may be key for overcoming the bias.

Our investigation revealed a robust sex bias in pain management. This is in line with another study that has also documented a sex bias in pain management (25), but not with a few other studies that did not find such a bias (28–30). We speculate that a previously overlooked factor may explain the presence or absence of a sex bias in

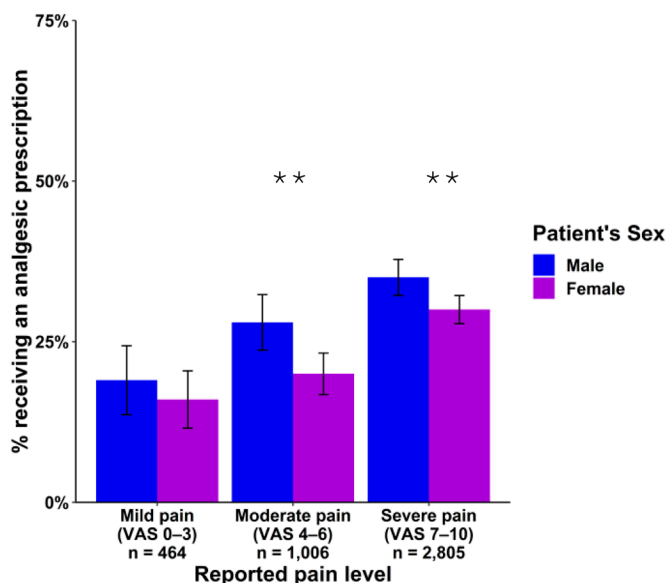


Fig. 3. Likelihood of receiving an analgesic prescription by patient's sex and reported pain level (VAS) (Study 2). Error bars denote 95% CI. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

these studies. Specifically, in studies that did not find a sex bias, many of the pain complaints had a clear physical source (such as trauma, fracture, or infection). In such cases, healthcare providers may determine the pain management based on the objective cause of the pain rather than on their perception of the pain report. By contrast, in studies (including the present one) that documented a sex bias, most pain complaints had a vague source, such as head or abdominal pain. Here, healthcare providers must heavily rely on their perception of the patient's pain, and so a biased perception is likely to lead to bias in pain management decisions. Future research should test whether a sex disparity is more likely to be found when pain complaints are more ambiguous and the source of the pain is unknown. This factor may join other factors modulating the sex bias, such as organizational culture. Sex bias may be less pronounced in cultures that prioritize sex and gender equality (46) or in health systems that integrate female-focused health training into the curriculum. We believe that a systematic meta-analysis, as well as new studies spanning different countries, cultures, and healthcare systems, could shed light on factors moderating the sex bias in pain management.

The present findings portray a somber account of the pain treatment that females receive at discharge from the ED. Healthcare providers in two countries appear more likely to doubt females' pain reports, they prescribe less medication to relieve females' pain, and they diverge more from medical guidelines. Inadequate pain management is known to lead to unnecessary suffering and deleterious health effects (2–4) and also carries preventable costs for public health (5). How may the healthcare system overcome this bias? One solution that has been identified in other medical contexts is sex concordance: providing treatment by same-sex healthcare providers (33, 34). Yet, sex concordance may not be effective if female and male healthcare providers are both subject to the same stereotypes and view females' pain reports as exaggerated. Indeed, we find that male and female physicians undertreat female patients to a similar extent, and nurses, who are mostly female, perceive females' pain in a biased manner and are less likely to record females' pain scores. Another potential way to overcome the sex bias may be through computerized decision-support tools nudging healthcare providers (47) to record a pain score for all patients and prescribe an analgesic according to the reported pain

score. Raising healthcare providers' awareness of sex bias and educating them on its harms may also help reduce it (48–50). Future research should explore these potential debiasing tools.

Our studies were limited to pain management decisions in the ED. The ED is a central location for pain treatment, but pain is treated in many other locations, from family clinics to post-operative settings. Furthermore, reactions and decisions based on pain assessment are not confined to the medical context and occur in common day-to-day settings. Further research should test whether sex bias is present in such settings. It should also be recalled that our analysis examined only sex-based disparities since gender identity was not recorded in our databases. It is crucial to investigate whether a gender-based bias also prevails in pain management (51).

To conclude, the present research provides robust evidence for healthcare providers' sex bias against female patients in pain management. We identify the stereotypical perception of females' pain as one of the potential mechanisms underlying the bias. The findings join mounting evidence of discrimination against females in the medical system and in other areas. Undertreatment of females' pain bears immediate implications for the healthcare system and broad implications for society's attitude toward female pain.

Methods

Study 1.

Data collection. The data included 17,576 individual discharge notes from the years 2014 to 2019. We aimed to examine discharge notes where the appropriate pain management was not straightforward. Consequently, we obtained discharge notes that fulfilled the following prespecified inclusion criteria: adult patient (18+ years old); having a pain-related diagnosis (e.g., headache, backache) and no diagnosis of trauma, neoplasm, poisoning, or cerebrovascular accident; discharged directly from the ED and not hospitalized. The data included personal characteristics of the patient: sex, age group, ethnicity, pregnancy status, time of admission and discharge from the ED, and whether the patient self-referred to the ED (*SI Appendix, Table S1A*). Fifty-four percent of the discharge notes were of female patients. The data also included patients' medical information: the primary pain diagnosis; whether there was a diagnosis of a chronic pain condition (e.g., chronic lower back pain); the reported pain score (VAS; reported on a 0 to 10 Visual Analogue Scale; recorded in ~40% of the cases); and the medications prescribed upon discharge. The data included variables related to the ED setting: number of patients in the ED at the time of discharge, type of ED, and campus. Finally, the data included variables related to the physicians (*SI Appendix, Table S1B*): sex, age, ethnicity, and whether or not the physician was in residency training. The data included 749 physicians, of which 31% were female. The study was approved by the Hadassah Medical Center IRB committee (Protocol no. 0563-20-HMO). Due to the deidentified nature of the data, informed consent was waived.

Statistical analysis. We compared analgesic prescriptions by patient's sex and pain level using a chi-square test for independence. We repeated these analyses for opioid analgesics and for nonopioid analgesics separately. We compared pain scores via a *t* test for independent samples. Next, we performed a hierarchical multivariate linear probability analysis where the unit observation was a unique visit, and SE were clustered at the physician level. In the first step, the only explanatory variable was the patient's sex. In the second step, the explanatory variables included patient and ED characteristics. In the third step, we added physician characteristics to the model. In the final step, we added the patient's sex by physician's sex interaction. Additional analysis tested separately several other interactions between patient's sex and patient, physician, and ED characteristics. We repeated the main regression analysis for patients who visited the ED only once. Next, because these regression analyses included only the cases with recorded pain scores, we repeated the main regression analyses for the full cohort and for those with no recorded pain scores. We conducted a similar multivariate linear probability analysis predicting triage nurses' recording of pain scores. Finally, we conducted a multivariable linear regression analysis predicting the length of stay at the ED. In all tests in this study and the next ones, we set an a priori two-sided

$\alpha = 0.05$. In this study and the next ones, analyses were conducted using R version 4.1.2 and SPSS version 25.

Study 2.

Data collection. The dataset included 4,275 discharge notes from 2019, covering all the cases fulfilling the same prespecified inclusion criteria as in Study 1. Fifty-nine percent of the discharge notes were of female patients. Due to IRB constraints, the data included only the aggregative analgesic prescription rates (with no specification of the type of analgesic) by the patient's sex and pain score. The study was exempted from IRB review and informed consent was waived.

Statistical analysis. We compared analgesic prescriptions by patient's sex and pain level using a chi-square test for independence. We compared pain scores via a *t* test for independent samples.

Study 3.

Study population. Practicing nurses and physicians at the University of Missouri Health Care Hospital were invited, through the hospital's mailing list, to voluntarily participate in an online survey ($n = 109$, 96% nurses, 85% females, $M_{age} = 40$, $SD_{age} = 13$).

Study plan. We preregistered our analyses and hypotheses at: https://aspredicted.org/TGD_DN6. The study was approved by the University of Missouri Health Care Hospital IRB committee (Case #2096131).

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Statistical analysis. We compared participants' judgments of a male and a female patient's pain via a *t* test for independent samples. The sample size did not allow further analyses by the participant's sex or profession.

Data, Materials, and Software Availability. Anonymized patient data have been deposited in <https://osf.io/heznx/> (52).

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